NEXTCYCLE
MICHIGAN

MICHIGAN 2025 BATTERY GAP ANALYSIS

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COMMISSIONED BY:

EGLE
MICHIGAN DEPARTMENT OF ENVIRONMENT, GREAT LAKES, AND ENERGY

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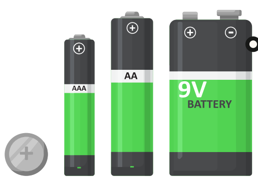
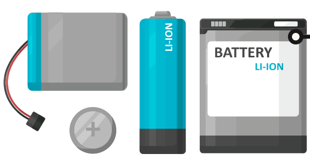
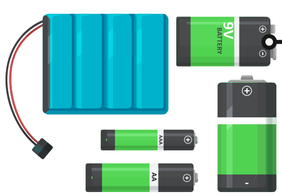
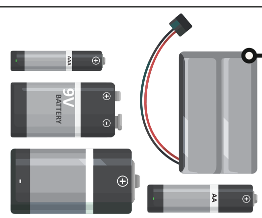
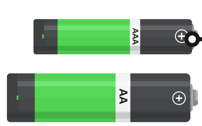
EXECUTIVE SUMMARY

Batteries are increasingly common and complex in consumer products, creating a growing challenge for Michigan's waste and recovery systems. Batteries also represent an important opportunity to recover critical minerals, reduce reliance on virgin resource extraction, and provide end markets with a valuable source of recycled feedstock. Expanding battery recovery and recycling can also strengthen domestic supply chains and reduce national security risks by lessening dependence on foreign sources of critical materials. Communities across the state are raising concerns about battery-related risks in the waste stream, underscoring the need for consistent public education and convenient recycling access. Recognizing these challenges, EGLE is leveraging U.S. DOE grant support to develop a comprehensive strategy for battery outreach, education, recycling, and reprocessing. This report represents the initial gap analysis, with further work planned to expand upon this topic.

Table 1 outlines the year one gap analysis battery type focus: small- and medium-format batteries. Included in the table are the battery types, common sizes and formats, as well as a select list of battery examples. This study does not include automotive lead-acid batteries, which are already prohibited from landfill disposal and are commonly returned to retailers for recycling or managed through household hazardous waste (HHW) facilities and scrap metal processors.

Table 1: Small and Medium Battery Formats included in the Gap Analysis Research

	BATTERY TYPE	COMMON FORMS	SELECT EXAMPLES
	Alkaline and Zinc-Carbon Batteries, Lithium Single-Use (Lithium "Primary")	AAA, AA, 9V, Button-Cell, Coin	Watches, musical cards and books, string lights, digital scales and thermometers, cameras, smoke detectors, handheld games, remote controls, hearing aids, medical devices, e-cigarettes/vapes, keyless vehicle entry remotes
	Lithium-Ion Batteries (Li-ion) (Lithium "Secondary") (Rechargeable)	Prismatic (flat), Cylindrical (e.g., 18650), Pouch, Button-cell	Cellphones, e-cigarettes/vapes, laptops, digital cameras, consumer electronics, some small and large household appliances (may not be removable.)
	Nickel-Cadmium Batteries (Ni-Cd) (Rechargeable)	AA, AAA, C, D, 9V, Pack (custom shapes for tools)	Cordless power tools, digital and video cameras, bio-medical equipment
	Nickel-Metal Hydride Batteries (Ni-MH) (Rechargeable)	AA, AAA, C, D, 9V, Pack	Older cellphones, digital cameras, cordless power tools, kids toys
	Nickel-Zinc Batteries (Ni-Zn) (Rechargeable)	AAA, AA	Consumer Electronics, Medical Devices

Note: The battery examples provided in this table are meant for illustrative purposes and are not intended as an exhaustive list.

BATTERY RECYCLING CHALLENGES

RESIDENTS

Confusion at Home

Many residents store used batteries at home or toss them in curbside trash or recycling bins due to uncertainty about proper disposal.

COMMUNICATION & ACCESS

Inconsistent Information

Guidance provided by local governments is incomplete and residents have limited access to drop-off sites and retail recycling programs.

PROGRAM CHALLENGES

Many Barriers to Success

Battery recycling is limited by contamination, unstable funding, high costs, low public awareness, and emerging risks.

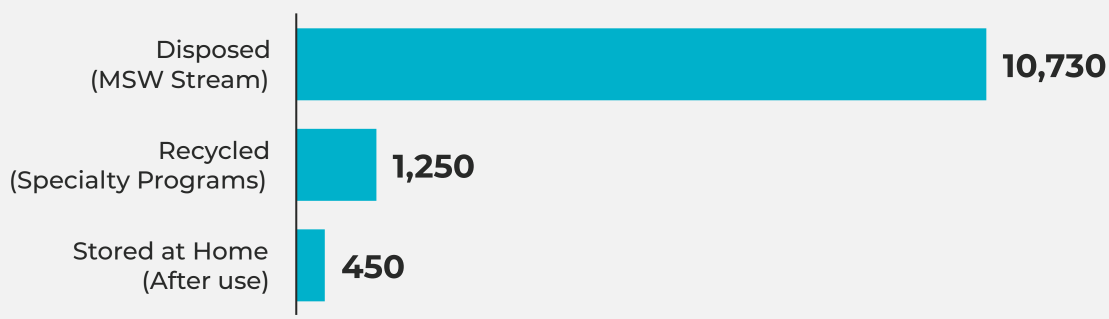
POLICY

Lack in Robust, Battery-Focused Policy

Extended Producer Responsibility (EPR) policies could ensure and fund stable, targeted, and comprehensive programs.

WHERE DO 12,400 TONS OF BATTERIES GO EACH YEAR?

ESTIMATED BATTERY MANAGEMENT METHOD (TONS)



WHAT WOULD IMPROVE BATTERY RECYCLING RATES?

54% of residents surveyed said **more convenient access** would encourage them to participate in battery recycling

- RETAIL PROGRAMS
- COMMUNITY DROP –OFF SITES
- LOCAL COLLECTION EVENTS

CONSUMER BATTERY SOLUTION RECOMMENDATIONS

ADVISORY GROUP	Establish an Advisory Group to guide battery recovery solutions and advance statewide strategies.
NEEDS ASSESSMENT	Assess existing recovery infrastructure and key challenges using the Battery Gap Analysis/Statewide Needs Assessment.
EPR LEGISLATION	Consider EPR legislation to shift costs to producers, align with other states, and secure stable funding. Consider having legislation plan for embedded and emerging batteries (e.g., e-mobility, vapes, electronics) based on a comprehensive statewide needs assessment.
PROGRAM DESIGN	Adopt design consistent with recent EPR laws to reduce opposition, simplify compliance, and support consistent consumer education; clarify retailer roles to avoid overburdening collection responsibilities.
ACCESS	Guarantee year-round, statewide access through convenient drop-off options in both urban and rural areas.
EDUCATION	Launch a statewide education and outreach campaign on battery recycling, leveraging catalyst communities and coordinated stakeholder engagement.
PROCESSING CAPACITY	Expand recovery and processing capacity through targeted infrastructure investments supported by EGLE, DOE, accelerators, and grant funding.

Key Report Takeaways and Action Items

CONSUMER BATTERY BEHAVIOR AND KNOWLEDGE

Resident challenges

Most residents face barriers to battery recycling, including confusion about proper recycling, limited convenient options, lack of clear information, and uncertainty that leads many to keep used batteries at home. While most residents know batteries should go to specialty recycling programs, many residents still believe batteries should go into curbside trash or recycling containers.

CONSUMER BATTERY GENERATION

High disposal volume

Due to confusion and limited recycling access, an estimated 10,700 tons of consumer batteries enter Michigan's MSW stream each year (≈86% of total generation), compared to just 1,250 tons recovered and 450 tons stored in homes after use.

MICHIGAN RESIDENTS ACCESS TO BATTERY RECYCLING

Gaps in communication and access

Local governments provide incomplete and inconsistent battery recycling information, fewer than one-third of residents have access to permanent drop-off sites, and many communities rely on voluntary retail programs that are limited and unstable.

CONSUMER BATTERY STAKEHOLDER INSIGHTS

Program challenges

Battery recycling is limited by contamination, unstable funding, high costs, and low public awareness, while emerging risks like vapes and increasing facility fires highlight the need for statewide education and consideration of expanded Extended Producer Responsibility (EPR) to ensure stable, comprehensive programs.

CONSUMER BATTERY POLICY REVIEW

Policy landscape

Michigan lacks a statewide consumer battery program, while Vermont and several other states (CA, WA, NJ, IL, NE, CT, CO) have adopted EPR laws that increasingly align on key provisions; following these models could streamline compliance, expand access, and provide a pathway for Michigan.

CONSUMER BATTERY SOLUTION RECOMMENDATIONS

EGLE could consider the following recommendations:

Advisory group

Establish an Advisory Group to guide battery recovery solutions and advance statewide strategies.

Needs assessment

Assess existing recovery infrastructure and key challenges using the Battery Gap Analysis/Statewide Needs Assessment.

EPR legislation

Advance EPR legislation to shift costs to producers, align with other states, and secure stable funding. Ensure that the legislation covers embedded and emerging batteries (e.g., e-mobility, vapes, electronics) based on a comprehensive statewide needs assessment.

Program design

Adopt design consistent with recent EPR laws to reduce opposition, simplify compliance, and support consistent consumer education; clarify retailer roles to avoid overburdening collection responsibilities.

Safety training

Expanding training for public sector staff and collection organizations, along with stronger public education and outreach on battery hazards, to reduce battery fire hazard risks.

Access

Guarantee year-round, statewide access through convenient drop-off options in both urban and rural areas.

Education

Launch a statewide education and outreach campaign on battery recycling, leveraging catalyst communities and coordinated stakeholder engagement.

Processing capacity

Expand recovery and processing capacity through targeted infrastructure investments supported by EGLE, DOE, accelerators, and grant funding.

TERMINOLOGY, DEFINITIONS, AND ACRONYMS

Acronyms

EGLE	Michigan Department of Environment, Great Lakes, and Energy
EPR	Extended Producer Responsibility
HHW	Household Hazardous Waste
MRC	Michigan Recycling Coalition
MSW	Municipal Solid Waste
U.S. DOE	United States Department of Energy

Definitions

E-Cigarette/Vape
E-cigarettes, also known as vapes, are battery-operated devices that heat a liquid and produce an aerosol, a mix of small particles released in the air.

Embedded Battery
A battery that is not designed to be easily removed by the user with no more than commonly used household tools, commonly found in electric toothbrushes.

Extended Producer Responsibility (EPR)
A policy approach that requires producers to take financial and/or physical responsibility for management of the products and/or packaging they produce at the end of their useful life.

Medium Battery Format Battery
A battery weighing between 11 to 25 pounds, commonly used in scooters, e-bikes and outdoor power equipment.

Municipal Solid Waste (MSW)
MSW refers to household waste, commercial waste, waste generated by other nonindustrial locations, waste with characteristics like that generated at a household or commercial business, or any combination thereof. MSW does not include municipal wastewater treatment sludges, industrial process wastes, automobile bodies, combustion ash, or construction and demolition debris (Act 451 1994).

Local Government Program
A program run by either a municipality, solid waste district, county, or specified communities.

Rechargeable/Secondary Lithium-Ion Batteries
Batteries commonly used in cellphones such as iPhone and Android, laptops such as Dell and Macs, digital cameras, consumer electronics, newer cordless power tools such as Ryobi, DeWalt, Milwaukee, rechargeable e-cigarettes/vapes.

Rechargeable/Secondary Nickel-Cadmium Batteries
Batteries commonly used in older cordless power tools, older digital cameras, and older bio-medical equipment.

Rechargeable/Secondary Nickle-Metal Hydride Batteries
Batteries commonly used in older type cellphones, digital cameras, cordless power tools, toys.

Rechargeable/Secondary Nickel-Zinc batteries
Batteries commonly used in older type consumer electronics and medical devices.

Single Use/Primary Batteries Alkaline and Zinc-Carbon Batteries
AAA, AA, C, D, 9V batteries commonly used in toys, consumer electronics, medical devices.

Single Use/Primary Lithium Batteries
Batteries commonly used in AAA, AA, 9V, button-cell or coin batteries commonly found in watches, musical cards and books, string lights, digital scales and thermometers, cameras, smoke detectors, handheld games, remote controls, hearing aids, medical devices, keyless vehicle entry remotes, non-rechargeable e-cigarettes/vapes.

Small Format Battery
A battery weighing less than 11 pounds.

INTRODUCTION

Batteries are part of an ever-growing array of consumer products. As both battery quantity and complexity increases in our products – from single use to rechargeable, loose and embedded – the need for consistent education and outreach to Michigan residents about proper battery recycling and convenient access to recycling options also grows. Many communities in Michigan are sounding the alarm about a growing battery problem in the waste and recycling streams. In May of 2025, a lithium battery caused a fire in a trash truck encouraging the Troy Fire Department to warn residents not to put these batteries in their trash or recycle bins (Waste360 Staff, 2025). In June 2025, The Grand Rapids Fire Department and Kent County Department of Public Works issued a warning to residents about the risk of fire from lithium batteries and noted that battery fires are occurring monthly at their recycling and waste facilities (MacLean, 2025). These examples are consistent with the 2025 Annual Waste & Recycling Facility Fires Report, which found an all-time high of 448 publicly reported fire incidents at waste and recycling facilities across the U.S. and Canada in 2025 (Fogelman, 2026). EGLE recognizes the challenges faced by local governments across Michigan to properly manage batteries in their waste streams. Building on alignment with federal priorities, EGLE is examining opportunities to develop a comprehensive strategy to enhance the outreach and education, recycling, and reprocessing of batteries in Michigan. Batteries represent an important opportunity to recover critical minerals, reduce reliance on virgin resource extraction, and provide end markets with a valuable source of recycled feedstock. Expanding battery recovery and recycling can also strengthen domestic supply chains and reduce national security risks by lessening dependence on foreign sources of critical materials. This report presents an initial gap analysis that will be expanded in subsequent phases of this work.

Table 2 outlines the gap analysis battery type focus: small- and medium-format battery. Included in the table are the battery types, common sizes and formats, as well as a select list of battery examples. This study does not include automotive lead-acid batteries, which are already prohibited from landfill disposal and are commonly returned to retailers for recycling or managed through household hazardous waste facilities and scrap metal processors.

Table 2: Small and Medium Battery Formats included in the Gap Analysis Research

BATTERY TYPE	COMMON FORMS	SELECT EXAMPLES
Alkaline and Zinc-Carbon Batteries, Lithium Single-Use (Lithium "Primary")	AAA, AA, 9V, Button-Cell, Coin	Watches, musical cards and books, string lights, digital scales and thermometers, cameras, smoke detectors, handheld games, remote controls, hearing aids, medical devices, e-cigarettes/vapes, keyless vehicle entry remotes
Lithium-Ion Batteries (Li-ion) (Lithium "Secondary") (Rechargeable)	Prismatic (flat), Cylindrical (e.g., 18650), Pouch, Button-cell	Cellphones, e-cigarettes/vapes, laptops, digital cameras, consumer electronics, some small and large household appliances (may not be removable.)
Nickel-Cadmium Batteries (Ni-Cd) (Rechargeable)	AA, AAA, C, D, 9V, Pack (custom shapes for tools)	Cordless power tools, digital and video cameras, bio-medical equipment
Nickel-Metal Hydride Batteries (Ni-MH) (Rechargeable)	AA, AAA, C, D, 9V, Pack	Older cellphones, digital cameras, cordless power tools, kids toys
Nickel-Zinc Batteries (Ni-Zn) (Rechargeable)	AAA, AA	Consumer Electronics, Medical Devices

Note: The battery examples provided in this table are meant for illustrative purposes and are not intended as an exhaustive list.

CONSUMER BATTERY BEHAVIOR AND KNOWLEDGE

Section Summary

This section summarizes the findings of a residential consumer battery survey. More than 770 responses were collected statewide, reflecting Michigan's demographic profile. The survey assessed residents' knowledge of battery recycling, the number of batteries in use and replaced annually, and how batteries are managed at end of life.

Key findings include:

- **Confusion about proper management:** About one-third of respondents believed batteries could be placed in curbside trash or recycling carts, while two-thirds recognized the need for dedicated recycling programs.
- **Difficulty recycling:** Only 5% of respondents reported no barriers to battery recycling. Common barriers included uncertainty about recycling methods for different battery types, lack of clarity on where to take batteries, and limited access to convenient drop-off options.
- **Need for more information:** Residents expressed interest in clearer guidance on safe handling, where to recycle batteries, and how to identify different battery types.
- **Unused batteries kept at home:** Nearly half of respondents who held onto replaced batteries said they did so because they were unsure how to properly dispose of or recycle them.

Residential Consumer Battery Survey

A critical gap in Michigan's battery recovery system is the lack of reliable data on how many consumer batteries residents generate each year and how they manage them at end of life. Without this information, it is difficult to assess the true need for recovery services or design effective collection programs. Key questions include how many batteries are currently stored in households, how many were removed in the past year, and whether those removals were managed through disposal, improper recycling channels, or battery specific recycling programs.

To address this data gap the Project Team conducted a statewide residential battery survey in summer 2025. The goals of the survey were to:

- Quantify number and type of consumer batteries stored in Michigan residents' homes and batteries replaced in the last year;
- Establish methods of managing replaced batteries used by residents;
- Gather understanding of residential knowledge and insights to consumer battery recycling across the state.

Over a three-week period, 776 responses were collected from residents in 344 unique ZIP codes, spanning all 14 Councils of Government. This geographic diversity provides confidence that the findings reflect battery behaviors and perceptions across the state.

Survey responses were representative of Michigan's demographic population in age, race, gender, and geographic distribution. 35% of responses came from rural counties, 62% from urban counties in Michigan's lower peninsula, and 3% of respondents came from Michigan's Upper Peninsula, in alignment with the state's population distribution.

The survey focused on the battery types listed in Table 2. To support accurate responses, images of each battery type were included to help respondents identify them correctly. Participants reported both the number of batteries stored in their homes and those removed in the past year, as well as the method of management and their understanding of proper recycling practices. Reported counts were then converted into weight estimates using average battery weights, allowing calculation of the total tons of batteries generated annually in Michigan by type and management pathway.

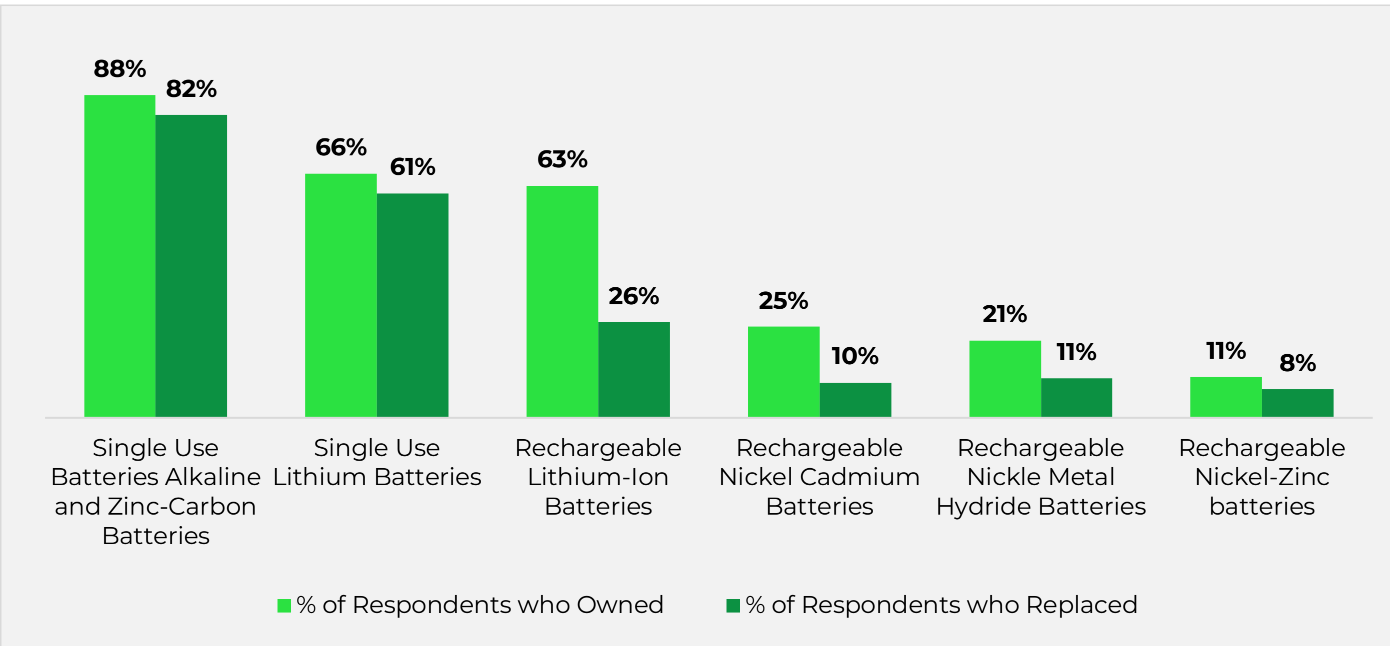
By filling this information gap, the survey provides Michigan with its first current, representative dataset on consumer battery generation. These findings are essential for planning effective recovery services, ensuring safe handling, and supporting future policy and program development.

Battery Ownership and Replacement

Figure 1 presents the percentage of respondents who reported which battery types they currently had in their home and which they replaced in the last year. Single-use batteries including alkaline, zinc-carbon, and single-use lithium were both widely owned and frequently replaced. 80% of respondents reported owning alkaline or zinc-carbon batteries, and 66% reported owning single-use lithium. Reported replacement rates were nearly identical: 82% for alkaline or zinc-carbon and 61% for single-use lithium. This near parity between ownership and replacement indicates that single-use batteries are turning over rapidly in households—purchased, used, and removed within the same year—consistent with their short lifespan.

Rechargeable batteries showed a very different pattern. Ownership rates were higher than replacement rates, suggesting that many rechargeable batteries remain in households for multiple years. Lithium-ion was by far the most dominant rechargeable type, with 63% of respondents reporting ownership compared to 26% who reported removing them in the past year. Other rechargeable chemistries including nickel-cadmium, nickel-metal hydride, and nickel-zinc were far less common, as illustrated in Figure 1. The prominence of lithium-ion is not surprising given its superior performance relative to older rechargeable technologies and its rapid replacement of those chemistries in the consumer marketplace.

Figure 1: Battery Ownership and Replacement Trends

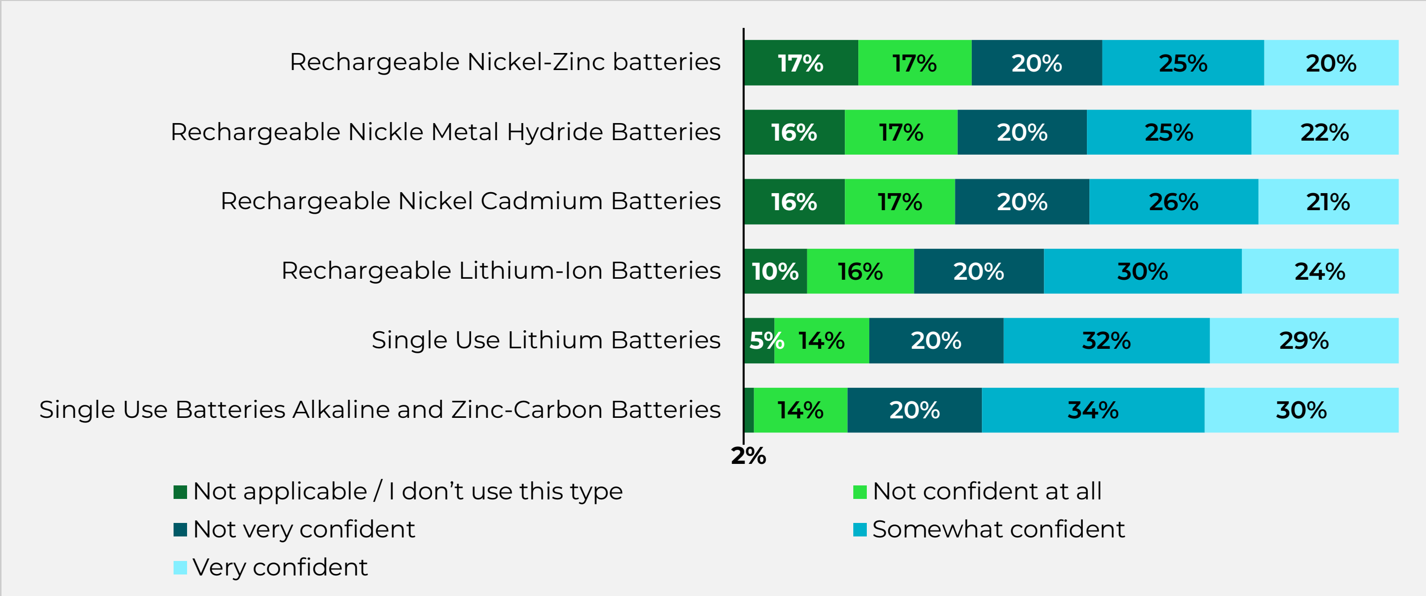


Knowledge and Awareness

Residents were asked to provide their perceived knowledge of proper battery management and gathered feedback on ways to improve battery recovery efforts across the state.

When asked how confident residents felt about their knowledge on how to properly manage batteries at the end of life, residents responded some or high confidence in how to properly manage batteries ranging from 45% for rechargeable nickel-zinc batteries up to 64% in the case of single use alkaline and zinc-carbon batteries (Figure 2).

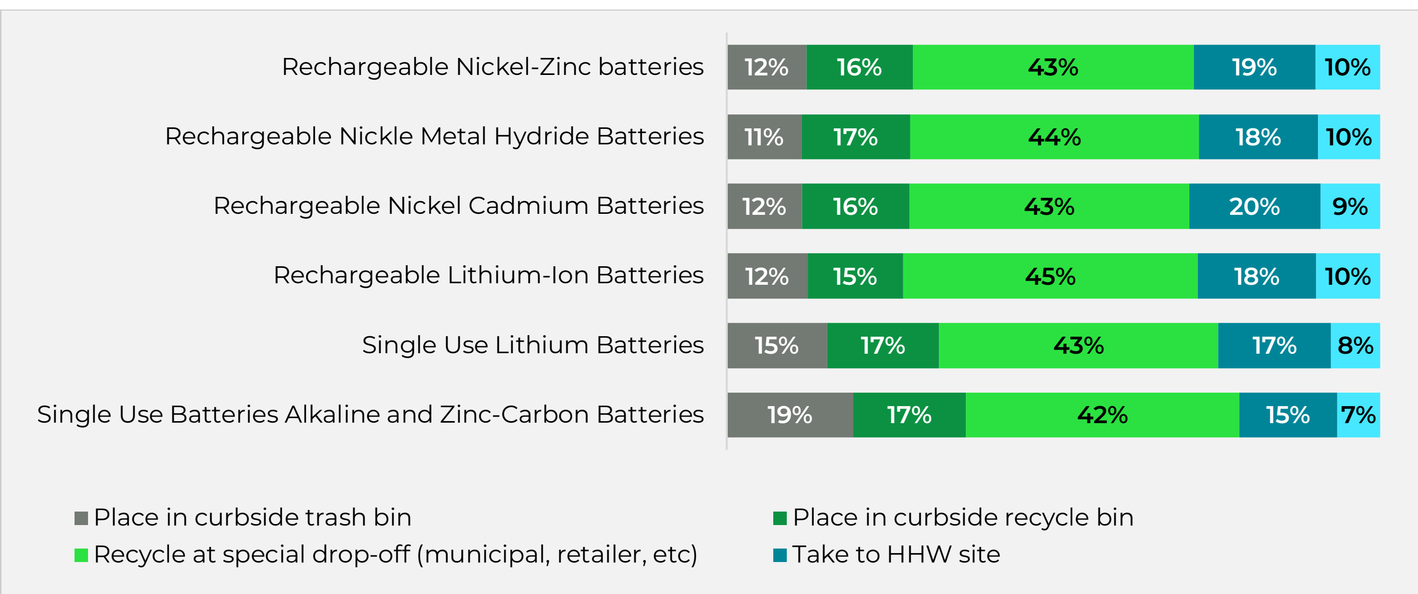
Figure 2: Residents Perceived Confidence in How to Properly Manage Batteries by Battery Type



At the same time, residents were asked what they believed to be the proper management method for different types of batteries (Figure 3). A notable share of respondents expressed misconceptions: between 15% and 17% indicated that batteries should go in the curbside recycling bin, and 11% to 19% believed they should be placed in curbside trash—perceptions that varied by battery type, with higher rates of curbside trash disposal for single-use batteries and lower rates for rechargeables. In contrast, most respondents recognized that batteries require specialized handling. For example, 63% indicated that single-use alkaline batteries should be processed through a dedicated recycling channel, and that number rose to 73% for lithium-ion batteries. Residents were also asked about best care practices for end-of-life batteries and battery-containing items. Most were familiar with general tips, such as storing batteries in a cool, dry place away from other conductive items. However, residents were less knowledgeable about more specific practices, including proper handling of vapes and e-cigarettes, as well as how to safely store damaged batteries to reduce fire risk.

These results highlight both encouraging awareness of the need for special management and the persistence of confusion that can lead to unsafe and ineffective disposal practices.

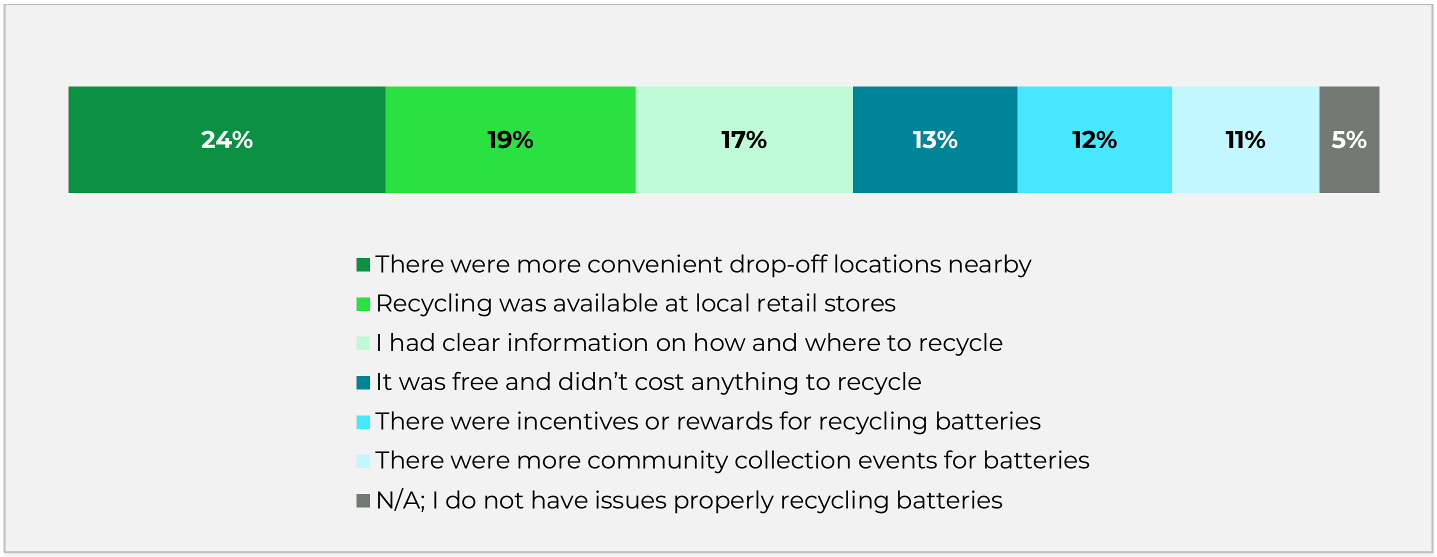
Figure 3: Residents Perceived Proper Management Strategy for Batteries by Battery Type



Resident Feedback

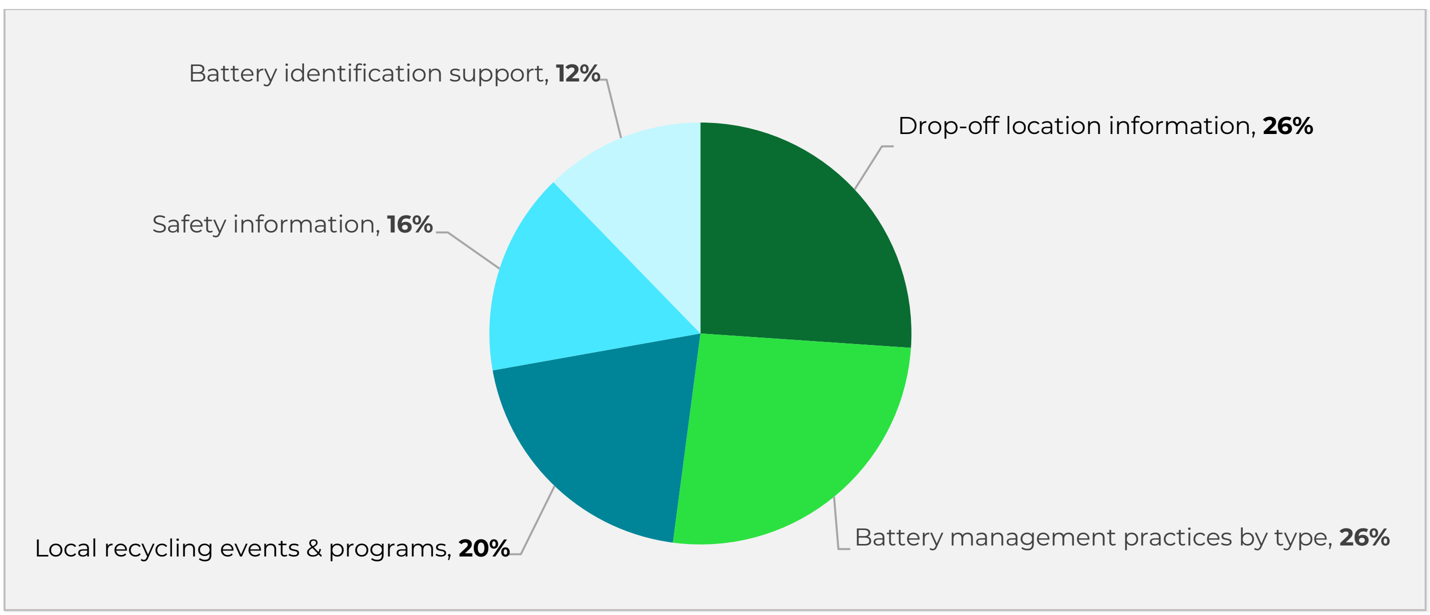
The survey found that 95% of residents face at least one barrier to recycling batteries, with only 5% reporting no obstacles. The most common challenges were uncertainty about where to take batteries, confusion over which types should be recycled, and limited local options. When asked what would encourage greater participation, residents emphasized the need for more convenient access—through retailers, community drop-off sites, or local collection events. About 17% also highlighted the lack of clear information on how and where to recycle, pointing to the need for both expanded access and stronger public education. (Figure 4).

Figure 4: Resident Motivations to Recycle Batteries



Looking ahead, residents expressed a desire for more community support. Specifically, they would like to see information on where to take batteries for recycling, as well as educational materials on different types of batteries and proper end-of-life management (Figure 5).

Figure 5: Resident Feedback on Desired Community Support



End-of-Life Management

When asked how they managed batteries replaced within the last year, respondents' habits revealed three key trends show in Figure 6.

- **Curbside Waste Services**
  - » Respondents reported that 32% of batteries were disposed of in the curbside trash
  - » Respondents reported that 11% of batteries were placed in curbside recycling
- **Battery Recycling Programs**
  - » Respondents reported that 13% of batteries were brought to a retail drop-off location
  - » Respondents reported that 12% of batteries were brought to a community or county drop-off program
  - » Respondents reported that 12% of batteries were brought to a HHW site
  - » Respondents reported that 4% of batteries were sent to a battery mail-back program
- **Kept at Home**
  - » Respondents reported that 17% of batteries were stored at home after replacement

Overall, the largest single reported trend was residents placing batteries in their curbside bins, reported by 42% of respondents. Nearly as many, 41%, reported using formal recycling programs such as retail or community drop-off, HHW facilities, or mail-back services. Another 17% reported that batteries were stored at home after replacement (Figure 6).

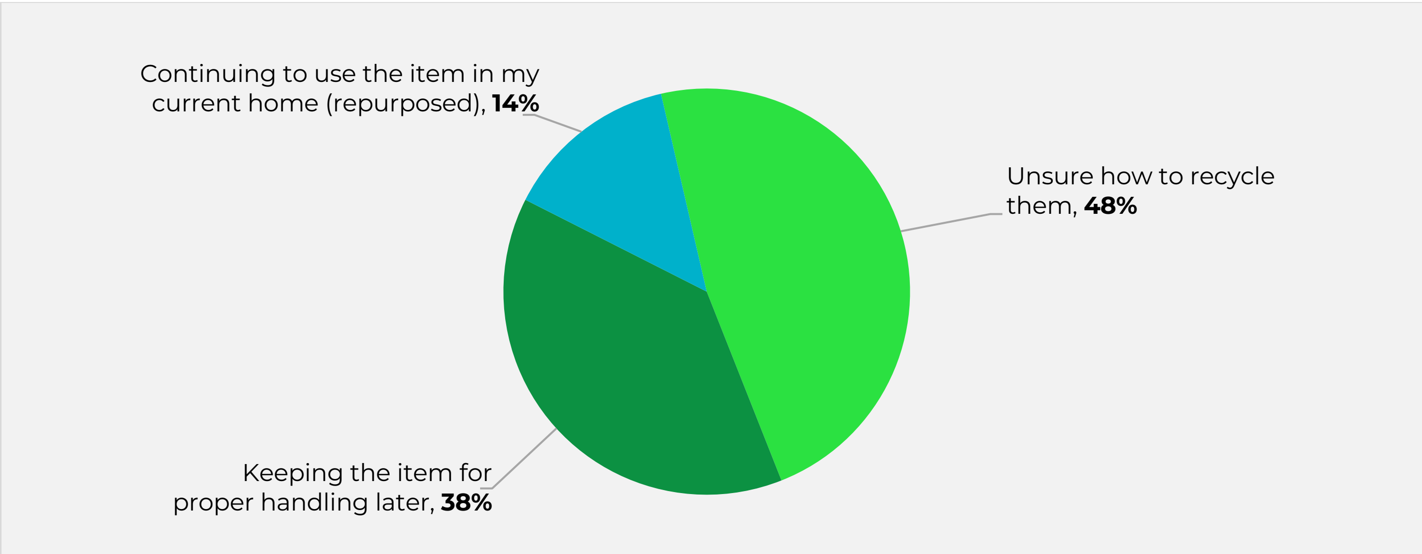
These findings highlight a critical issue: placing batteries in curbside trash or recycling bins poses both safety and recovery challenges. Batteries, especially lithium-ion batteries, can spark fires in collection trucks or at material recovery facilities, putting workers and infrastructure at risk. In addition, once mixed with curbside waste streams, batteries cannot be effectively recovered, representing both a safety hazard and a loss of valuable materials. Residents that place batteries in curbside recycling services may believe or wish those items to be recovered and unaware that their actions are causing more harm than good. Expanding safe, accessible recycling options and educating residents on proper management is therefore essential.

Figure 6: Residential Survey Responses for End-of-Life Management for Batteries

Kept at Home

Residents who kept batteries no longer in use at home were asked to provide additional context. The most common reason, cited by 48% of these respondents, was uncertainty about how to recycle batteries properly which was noted across all battery types in the survey. Another 38% stated they were holding onto batteries while waiting to dispose of them through mail-back or drop-off programs, and 14% indicated they were repurposing the replaced batteries within their homes (Figure 7).

Figure 7: Respondents Reason for Keeping Batteries No Longer in Use in the Home



End-of-Life Management Survey Responses Compared to State Waste Characterization Study

A comparison of residential survey responses on battery disposal with recent waste characterization studies reveals that survey respondents significantly undercount the number of batteries disposed.

The 2024 Michigan Municipal Solid Waste (MSW) Characterization study estimated that at least 29 million lithium-ion batteries are discarded annually, based on extrapolation from the number of e-cigarettes/vapes found in the waste stream. By contrast, survey responses, covering both single-use and rechargeable lithium-ion batteries, suggested only 6.9 million batteries are disposed.

Similar patterns appear in five other state waste characterization studies that directly sorted batteries from the MSW stream, all showing that surveys substantially underestimate actual disposal. Table 3 presents the waste characterization studies drawn on for extrapolating battery estimates to Michigan's population, and Figure 8 compares survey-based estimates of primary batteries discarded in trash or recycling bins with the extrapolated waste characterization estimates. Even the lowest study-based estimates, Michigan 2024 and Oregon 2024, both of which only identified vapes, suggest 29 to 42 million vapes alone are disposed annually in Michigan. The highest estimate, based on extrapolating findings from a New York state waste characterization study to the population of Michigan, places the estimate of primary batteries as high as 416 million.

A similar comparison for rechargeable batteries is shown in Figure 9. Fewer studies have examined the presence of rechargeable batteries in the disposal stream, however based on the waste characterization studies rechargeable batteries could vary from none to 86 million in Michigan's MSW disposal stream.

Table 3: State Waste Characterization Studies Used for Michigan Estimate

EXTRAPOLATION NUMBER	WASTE CHARACTERIZATION STUDY USED FOR EXTRAPOLATION TO MICHIGAN	NOTES
	Michigan Waste Sort	Vapes Only
1	Washington 2020	
2	Oregon 2024	Vapes Only
3	Vermont 2023	
4	New Hampshire 2024	
5	New York 2024	

Note: Not all states sorted for rechargeable batteries.

Figure 8: Primary Battery Disposal Estimates in Michigan

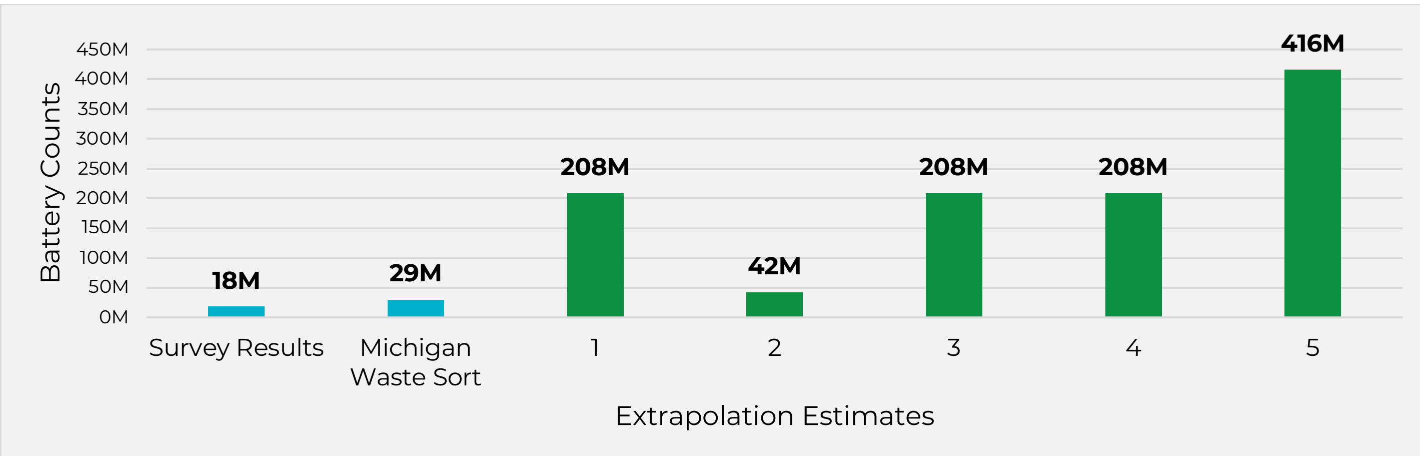
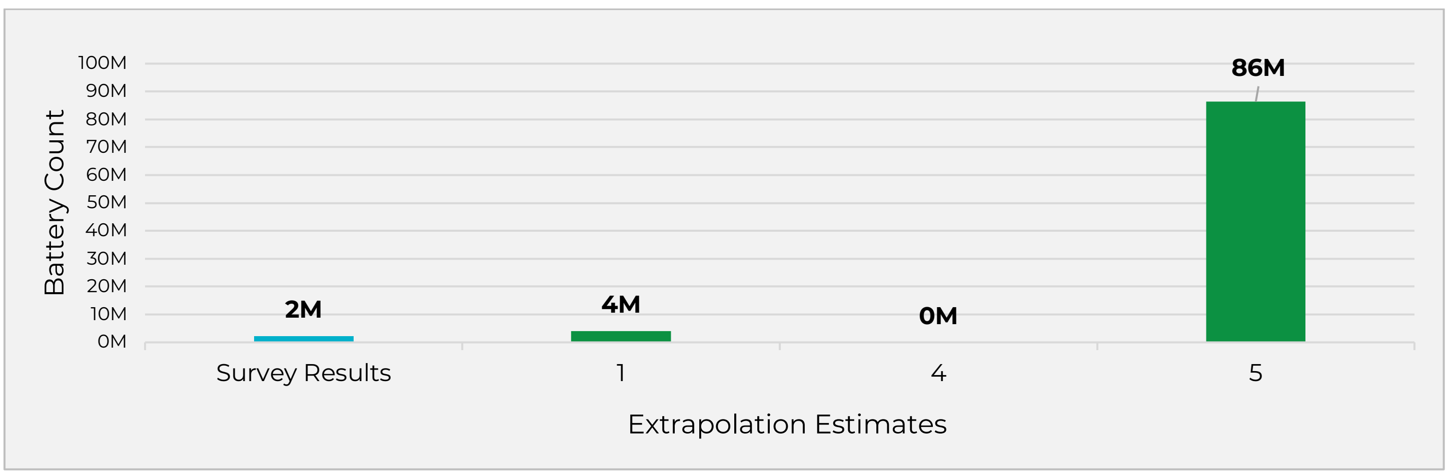


Figure 9: Rechargeable Battery Disposal Estimates in Michigan[1]



The undercounting of disposed batteries by survey respondents compared to waste characterization studies likely reveals a recall bias in the survey results. Residents may not recall all batteries they threw away and once out of sight, the batteries are out of mind. Additionally, batteries can be embedded in products so that consumers may not even be aware of the batteries being disposed.

[1] The Michigan waste characterization study did not examine the volume of rechargeable batteries in the waste stream.

CONSUMER BATTERY GENERATION

Section Summary

This section summarizes the estimated annual consumer battery generation rate in Michigan.

Key findings of this section include:

- **Extrapolation methodology:** The survey helped estimate the number of batteries retained at home after use and recycled through special drop-off programs. Waste characterization studies provided the baseline for estimating overall consumer battery disposal in Michigan.
- **High disposal volume:** An estimated 10,700 tons of consumer batteries enter Michigan's MSW stream each year, representing about 86% of total generation. In comparison, only 1,250 tons are recovered annually, while another 450 tons remain stored in homes after use.

Generation Estimate Methodology

A key component of the first-year consumer battery gap analysis is an estimate of annual generation in Michigan. The survey results, along with extrapolations from the waste characterization studies, are the core sets of data to estimate the extrapolation. The source data for each extrapolation component is described below:

- **Kept at Home**
  - » Residents were asked to take the survey at home with the ability to check and retrieve batteries directly.
  - » Because residents had access to the batteries during the survey, results are less prone to recall bias.
  - » Estimates of batteries kept in the home can only be obtained via survey responses.
- **Recycled**
  - » Residents who took extra steps to recycle are more likely to accurately recall the number and type of batteries recycled.
  - » Survey results indicate higher recycling activity than The Battery Network's (formerly Call2Recycle) reported recovery estimates for Michigan which is expected as The Battery Network is only one entity recycling batteries.
  - » Because no complete recycling dataset exists, survey responses serve as the best available proxy.
  - » However, self-reported recycling data may still be subject to recall bias.
- **Disposed**
  - » Survey responses significantly undercount batteries disposed, as shown by comparisons with waste characterization studies.

This undercount reflects recall bias, especially for routine disposal where little attention is paid. Waste characterization studies were used to estimate total consumer battery disposal.

Survey responses were extrapolated to estimate total battery generation in Michigan using average battery weight estimates and Michigan Census population data. This analysis provides insight into both the total stock of batteries currently in households and the annual volume recycled by residents. It should also be noted that contamination can still occur in recycling streams. A resident reporting that they "recycled" batteries does not guarantee proper preparation or sorting. For example, residents that reported bringing batteries to a drop-off may not have brought the batteries to the correct drop-off site or placed in the proper bin.

To estimate disposal, average proportions from waste characterization studies were applied: 0.13% of total waste by weight for primary batteries and 0.14% for rechargeable batteries, multiplied by the total MSW disposed of in Michigan in 2024.

Total generation estimates were then compared to consumer battery sales data.

Michigan Battery Generation Estimate

An estimated 12,430 tons of consumer batteries are replaced annually in Michigan with an estimated 10,730 tons entering the disposal stream, 1,250 tons entering recycling channels, and 450 tons of consumer batteries kept at home though no longer in use (Table 4 and Figure 10). Overall, 86% of batteries coming out of use annually are entering the disposal stream, either directly through residents throwing the batteries in curbside trash collection containers or through "wish-cycling" or as contamination in recycling streams.

Table 5 presents the breakdown of battery management method by battery type.

Table 4: Estimated Battery Management Method (Tons)

BATTERY MANAGEMENT METHOD	ESTIMATED TONS	PROPORTION
Kept at Home	450	4%
Recycled	1,250	10%
Disposed	10,730	86%
Total	**12,430**	**100%**

Figure 10: Estimated Battery Management Method (Tons)

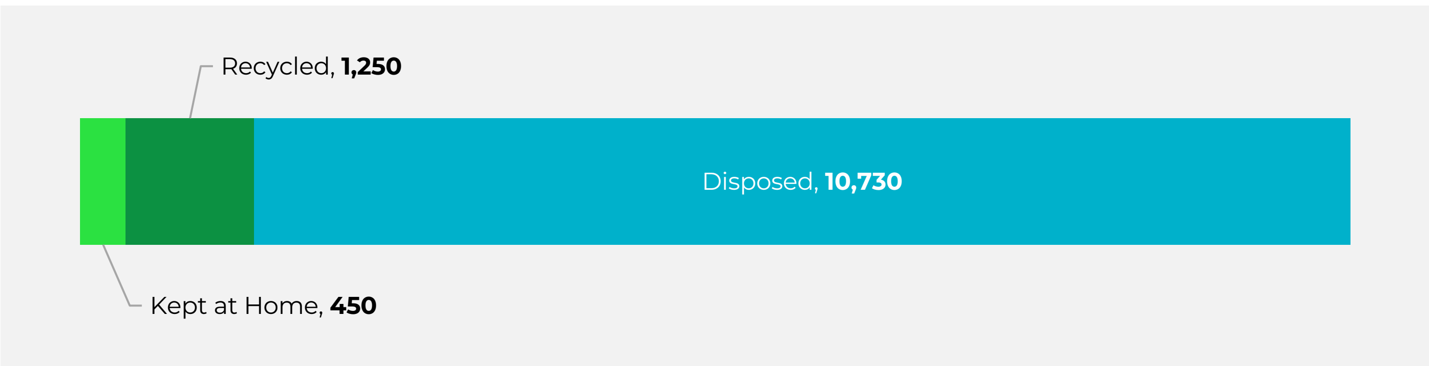


Table 5: Detailed Estimate Consumer Battery Management Methods (Tons)[2]

		Single Use Batteries Alkaline and Zinc-Carbon Batteries	Single Use Lithium Batteries	Rechargeable Lithium-Ion Batteries	Rechargeable Nickel Cadmium Batteries	Rechargeable Nickle Metal Hydride Batteries	Rechargeable Nickel-Zinc batteries	Total	Proportion
Kept – 4%	Kept at Home	137	14	83	174	30	11	**449**	**4%**
Entered Recovery Stream - 10%	Local Retailer	73	14	86	242	37	4	**456**	**4%**
	Mail-Back Recycle Program	16	4	44	97	18	3	**181**	**1%**
	Community Recycling Drop-off Center	76	13	74	123	39	7	**331**	**3%**
	HHW Collection Site	88	13	47	114	21	4	**286**	**2%**
Entered Disposal Stream – 86%	Curbside Recycle Bin	1,461	209	447	1,180	199	39	**3,535**	**28%**
	Curbside Trash Bin	2,973	425	910	2,401	405	79	**7,192**	**58%**
	Total	**4,823**	**692**	**1,693**	**4,330**	**748**	**145**	**12,430**	**100%**

[2] Sums may vary by +/- 3-5 due to rounding

MICHIGAN RESIDENTS ACCESS TO BATTERY RECYCLING

Section Summary

This section summarizes local government communication about consumer battery recycling and residential access to programs.

Key findings include:

- **Incomplete information:** Most residents can find some battery recycling guidance on local government websites, but it often covers only certain battery types, leaving significant gaps.
- **Inconsistent messaging:** Local governments vary widely in how they describe battery types and management methods, creating a patchwork of inconsistent information across the state.
- **Inadequate access:** Fewer than one-third of residents have access to permanent government-operated drop-off sites, and these sites do not always accept all battery types. Many residents face limited, inconvenient, or no access to proper recycling options.
- **Private business reliance:** In some areas, private retailers provide the only recycling option. While valuable, these programs are voluntary, vary in the information they share and battery types they accept, and programs may end with little or no notice.

The residential consumer battery survey highlights a clear need to expand convenient access to battery recycling while also strengthening education and outreach to ensure residents know how to properly manage batteries at end of life. To better understand access across the state, the Project Team conducted two in-depth research efforts: 1) a review of how municipalities communicate battery recycling information on their websites, and 2) an assessment of the availability of online battery recycling information from private-sector retail collection programs.

Local Government Communications on Battery Recycling

To assess consumer battery recycling access across Michigan, local government communication to residents was reviewed, covering nearly 80% of the state's population. Programs that explicitly mentioned battery acceptance—such as specialty battery sites, HHW or e-waste events that included batteries, or other programs clearly identifying batteries—were counted as providing some level of access to residents. Additionally, programs not directly operated by a municipality or district but promoted on local government websites were categorized as third-party programs (e.g. retail take-back programs promoted on a municipal website). Each program was evaluated for battery types accepted, fee structure, whether the program operated permanent sites or periodic events, and whether a third-party program (such as The Battery Network retail locations) was identified on the program's website. While this research did not cover every community in the state, it reflects the experience of most Michigan residents seeking battery recycling information through their municipal websites.

70% of Michigan residents can find information on battery recycling on their local government's website. 9% of residents cannot find any information on battery recycling from local government websites but may still have access to battery recycling through private businesses. The remaining 20% of the state's population was not included in this review, however many of these residents are in rural areas, where program information is often less consistently communicated.

The most common form of access provided by local governments is through county or municipal program serving only their jurisdiction (58%). Another 7% of Michigan residents have access through a solid waste district program. 3% of residents can access a program that requires an individual permit or enrollment, while 2% of residents are directed to third-party programs such as private-sector retail take-back by their local governments.

Table 6: Battery Drop-Off Programs by Service Type

BATTERY DROP-OFF SERVICE TYPE	POPULATION WITH ACCESS	PERCENT OF STATE POPULATION REPRESENTED[3]
Resident-Only (County or Municipal Program)	5,891,216	58%
District-Only (Residents of Waste District)	671,142	7%
Other (Permit-Based or Individual Program)	299,210	3%
No Program Information Found	885,758	9%
Third Party Only	247,539	2%
Total	**7,994,865**	**79%**

Although 70% of residents can find some information on consumer battery recycling through their local government webpages, the information provided is often incomplete. Table 7 shows the share of the population that can locate recycling guidance by battery type, and in every case the percentages are lower than 70%. This indicates that local communication rarely covers the full range of consumer battery types. For example, only 52% of residents can find information on how to recycle alkaline and zinc-carbon batteries.

The presence of communication or even a program does not guarantee adequate information or access that meets all residents' battery recovery needs. Municipalities vary widely in how they describe proper battery management: some reference only alkaline batteries, others focus solely on rechargeables, while few mention both. This inconsistency results in uneven and incomplete recycling guidance across the state.

More importantly, the lack of a standardized approach means residents encounter different and sometimes conflicting information depending on where they live. This creates confusion, undermines public trust, and limits residents' ability to manage batteries responsibly.

Table 7: Population with Drop-Off Access by Battery Type

BATTERY TYPE	POPULATION WITH DROP-OFF ACCESS	PERCENT OF STATE POPULATION REPRESENTED[3]
Alkaline Zinc-Carbon	5,216,119	52%
Lithium Ion (Single-Use)	6,227,785	62%
Nickel Cadmium	6,313,115	63%
Nickel Metal Hydride	5,892,628	58%
Nickle Zinc	5,892,628	58%
Lithium Ion (Rechargeable)	6,789,919	67%

Another key factor in access is convenience. Only 29% of Michigan residents have access to a permanent battery recycling drop-off site operated by their local government, generally considered the most convenient and stable option (Table 8). An additional 39% have access through temporary collection events. While these events are important access points, their usefulness depends on frequency, scheduling (e.g., evenings or weekends), registration requirements, and the quality of communication to residents.

A small share (2%) of residents are directed by their local governments to private businesses offering battery drop-off. While promoting these options is valuable, relying solely on private businesses poses challenges: local governments cannot control which batteries are accepted, how materials are managed, or whether the site remains available, since participation is at the business's discretion.

Overall, Michigan residents have limited convenient access to battery recycling, highlighting the need to expand permanent and well-structured options statewide.

[3] The remaining 20% of Michigan state population was not researched as part of this work.

Table 8: Access by Drop-Off Setting

DROP-OFF SETTING	POPULATION WITH ACCESS	PERCENT OF STATE POPULATION REPRESENTED[1]
Local Government Permanent Site & Temporary Event, and Third-Party Business	154,313	2%
Local Government Permanent Site & Temporary Event	394,072	4%
Local Government Permanent Site Only	1,141,401	11%
Local Government Permanent Site and Third-Party Business	1,244,223	12%
Local Government Temporary Event and Third-Party Business	87,535	1%
Local Government Temporary Event Only	3,840,024	38%
Private Business Only	247,539	2%
No Program Information Found	885,758	9%
Total	**7,994,865**	**79%**

A final aspect of the access analysis examined whether residents were required to pay a fee to participate in a recycling program. Approximately 64% of Michigan's population has access to free battery drop-off through either a permanent site or a temporary event (Table 9). However, this figure reflects programs that may only accept one or two of the battery types included in this study, rather than the full range of consumer batteries. Other programs charged either a per-vehicle fee or a permit fee, typically ranging from $10 to $30 per visit, with costs varying by community. Third-party businesses also frequently applied fees, often based on the weight of materials, though in some cases no fee information was publicly available.

Table 9: Battery Drop-Off Payment Structure

PAYMENT STRUCTURE	POPULATION WITH ACCESS	PERCENT OF STATE POPULATION
Free	6,447,403	64%
Vehicle or Permit Fee ($10-$30)	458,417	5%
Third Party Business/Organization (Fee Varies)	203,287	2%

Several programs direct residents to third-party services for battery recycling (Table 10). Approximately 23% of the population were directed by their program operator to a third-party drop-off, with 21% covered by multiple options that included both third-party programs and county, waste district, or other providers. As shown in Table 10, 2% of the researched population lives in an area where a third-party drop-off was the only option mentioned on the program website.

It is important to note that Table 10 does not suggest that 23% of residents only have access to third-party programs. Rather, it tracks the proportion of the researched population where municipalities or counties actively incorporate third-party details into their communication about proper battery management. In reality, many more residents likely find private-sector drop-off options through independent web searches, a topic explored further in the next section.

One concern with program operators directing residents to third-party programs is that these services can change abruptly, often without notice. Unless the municipality has a direct working relationship with the provider or regularly updates its website, residents risk receiving outdated or incorrect information, leading to confusion, frustration, and erosion of public trust.

Table 10: Third-Party Access by Population

THIRD-PARTY ACCESS TYPE	POPULATION WITH ACCESS	PERCENT OF STATE POPULATION
Only Third-Party Access	247,539	2%
Third-Party Access with Other Service Options	2,105,472	21%
Total with Third-Party Access	2,353,011	23%

Private Sector Battery Recycling Access

Michigan residents have a variety of options for battery recycling through retail drop-off programs, though coverage and accepted materials differ significantly by chain and geography. The statewide locator supported by The Battery Network provides the most comprehensive single resource, listing alkaline, rechargeable, lithium-ion, and button cell options across multiple retailers.

Among individual companies, Batteries Plus and Battery Giant stand out for their relatively broad acceptance and significant presence in Michigan, with each operating around 10 stores. Both accept alkaline (some retailers charge by the pound for alkaline drop off), rechargeable, and lithium-ion batteries, however these retailers have the most limited locations (18 total) and are only available in the more populated areas of the state.

Large home improvement stores such as Home Depot and Lowe's (about 70 and 45 Michigan locations, respectively) also offer convenient access for many residents. However, their programs are narrower in scope: both chains accept rechargeable and lithium-ion batteries but do not collect alkaline or button cell. Their websites emphasize safe disposal but are less detailed than battery-specific retailers. Ace Hardware, with roughly 225 Michigan locations, also represents a major potential access point, but its recycling program is less transparent—its website offers little clarity, and accepted materials vary from store to store (Table 11).

Other major retailers, including Meijer, Target, and Walmart, do not appear to offer meaningful recycling options for household batteries, leaving large gaps in accessibility given their statewide presence. Electronics-focused stores like Best Buy have well-known recycling programs for items such as phones and appliances but do not consistently advertise battery collection. Staples stands out as a partial exception, providing drop-off opportunities for rechargeable batteries at many locations.

In full, accessibility is relatively high given the number of participating locations statewide, though significant gaps remain in what materials are accepted (particularly alkaline and single-use button cell, as retailers either don't accept or charge for recycling these types of batteries) and geography is often key, as rural areas typically have far fewer disposal options. The Battery Network locator remains an especially valuable tool, offering the most reliable statewide resource for identifying which stores accept which types of batteries.

Table 11: Battery Acceptance by Retail Location[4]

RETAILER	MI LOCATIONS	ALKALINE ZINC-CARBON	LITHIUM (SINGLE-USE)	LITHIUM ION (RECHARGEABLE)	NICKEL CADMIUM (RECHARGEABLE)	NICKEL METAL HYDRIDE (RECHARGEABLE)	NICKEL ZINC (RECHARGEABLE)
Batteries Plus	8	Yes	Yes	Yes	Yes	Yes	Yes
Battery Giant	10	Yes	Yes	Yes	Yes	Yes	Yes
Home Depot	70	No	Yes	Yes	Yes	Yes	Yes
Lowe's	45	No	Yes	Yes	Yes	Yes	Yes
Ace Hardware	225	Unknown	Unknown	Unknown	Unknown	Unknown	Unknown
Menards	33	No	No	No	No	No	No
Staples	25	Yes	No	No	Yes	Yes	Yes
Best Buy	31	No	No	No	No	No	No
Target	54	No	No	No	No	No	No
Walmart	117	No	No	No	No	No	No
Meijer	125	No	No	No	No	No	No

In summary, there is more work to be done in Michigan as the communication to residents between community programs and private retailers is inconsistent, and solution sets are needed to provide education for residents and funding for programs that cost retailers and communities money.

[4] In some instances, certain stores may have battery drop-offs that are not listed on the national website for the companies. For example, a retailer that partners with a municipality for a drop-off. However, these details were not found during research.

CONSUMER BATTERY STAKEHOLDER INSIGHTS

Section Summary

This section summarizes consumer battery stakeholder insights from interviewed service providers and local governments.

Key findings include:

- **Collection challenges**: Contamination in battery collection containers is common, and compliance with safety protocols varies.
- **Barriers to growth:** High operating costs, unstable funding, and limited consumer education restrict the expansion of battery recycling.
- **Growing risks:** Vapes and e-cigarettes are increasingly showing up in bins, posing costly and complex disposal challenges.
- **Needed support:** Interviewees called for a statewide education campaign and expanded EPR to provide stable funding and to cover all battery types.

Service Provider Interview Summary

Several service providers were interviewed for this work to provide their perspective on consumer battery recycling in Michigan. Below is a summary of the largest battery service providers in Michigan that were interviewed along with the key takeaways learned from those interviews.

THE BATTERY NETWORK (FORMERLY CALL2RECYCLE) (HTTPS://BATTERYNETWORK.ORG/)

The Battery Network operates a nationwide battery collection and recycling network whose scope varies depending on state law and program type. The organization describes itself as having evolved from a collection program into a broader national system providing collection, logistics, education, and compliance support. Its standard collection kits accept most household dry-cell batteries up to 11 pounds each, while higher-energy lithium-ion batteries over 300 watt-hours are handled through a separate high-energy battery program. In states with battery stewardship laws, The Battery Network may also operate approved programs covering both rechargeable and primary (single-use) batteries, depending on the legal requirements in that state.

The Battery Network operates a rechargeable battery collection program in Michigan through retail and other collection partners. Its website identifies national retail partners such as The Home Depot and Lowe's, and company materials also reference Staples and Best Buy as examples of drop-off locations consumers can find through its locator. For collection partners, the program provides official collection boxes with prepaid, pre-addressed return shipping labels, and the organization describes its service model as a convenient, turnkey collection and recycling option. Partnered national retailers generally qualify for free collection kits, while other sites may need to purchase approved boxes or otherwise qualify under program rules.

In Michigan, collected batteries first go to a third-party sorter where they are sorted by chemistry before being sent to appropriate downstream processors. After sorting, batteries are distributed to 15 approved downstream processors across the United States, with some processors handling multiple chemistries and others specializing in single streams.

The processing network has grown significantly since 2023 to accommodate increasing battery collection volumes, expanding by approximately 25% according to interview feedback from The Battery Network. In 2024, The Battery Network reported recycling 122,781 pounds of primary and secondary batteries in Michigan.

ERG (HTTPS://ERGENVIRONMENTAL.COM/)

ERG Environmental Services, operating across Michigan, Indiana, and Ohio, provides comprehensive waste management services for HHW, non-hazardous waste, universal waste, and electronic waste. In Michigan, ERG operates a permanent HHW drop-off facility open to the public in Livonia, where batteries are accepted alongside other HHW and electronic waste materials for a fee. ERG also provides collection, waste characterization, packaging, labeling, manifesting, transportation coordination, and related compliance services for households, communities, and other generators.

CIRBA (HTTPS://WWW.CIRBASOLUTIONS.COM/)

Cirba Solutions is a battery recycling materials and management company that provides collection, logistics, sorting, processing, and critical-mineral recovery services for end-of-life batteries. The company was first started in Michigan 35 years ago and has since expanded across six states. In Michigan, Cirba Solutions has a facility in Wixom where batteries are sorted by chemistry before being prepared for shipment to recycling facilities. In 2024, Cirba Solutions processed more than 10M pounds of batteries through their Wixom facility.

Below are the key findings from the battery service provider interviews:

- ***Safety concerns***
  Batteries pose a significant fire risk across collection, disposal, and recycling operations. Lithium-ion batteries are of particular concern because they are both highly reactive and an increasing share of the waste stream. Safe collection and transport require proper segregation, handling, storage, and procedures to prevent damage or hazardous discharge.

- ***Contamination issues***
  Collection containers often include non-battery items and improperly prepared batteries, creating safety and operational challenges. Service providers may require training, screening, and removal of contaminants before shipment, but compliance can vary across collection sites.

- ***Funding challenge***
  Expansion of consumer battery collection programs remains constrained without dedicated funding mechanisms such as EPR legislation. Under current funding structures, providers may have limited ability to broaden collection coverage, add participating sites, or increase volumes beyond the battery types already supported.

Future expansion for battery collection and processing in Michigan may require EPR legislation that mandates all batteries be collected and provides funding for increased accessibility, with grant programs offering alternative support mechanisms in states without EPR laws.

Municipalities

As part of the recycling access interviews the Project Team interviewed seven counties and waste authorities across Michigan representing small rural, rural and high density populations in the lower and upper peninsulas to gather information on consumer battery recycling access through municipal or private sector programs. Counties and solid waste districts interviewed included Delta, Emmet, Kent, Oakland and Washtenaw Counties, the Resource Recovery and Recycling Authority of Southwest Oakland County (RRRASOC), and the Van Buren Conservation District.

Some of the questions asked during the interviews included:

- Are you aware of opportunities for residents to recycle the types of batteries listed in the table above in your county?
- Can you describe battery recycling programs in your county for the batteries listed in the table?
- Are there any public works or private drop-offs? What types of batteries are accepted at the different drop-off locations?
- Are the drop-off programs year-round, seasonal, or one-day events?
- Has access to battery recycling changed in your county in the past 3 years? If so, can you please describe any of the changes?
- What is the biggest barrier your county or communities in your county face in expanding battery recycling access?
- How could EGLE support battery recycling in your county?

Below are the key findings from the local government interviews:

- ***Program structures vary***
  Each local government had its own approach to battery collection, ranging from high-convenience features like year-round and free service to more limited models that charge fees, require appointments, or operate only on specific days.

- ***Barriers to expansion***
  The biggest challenges are high operating costs for drop-off programs and the need for stronger consumer education about battery hazards.

- ***Emerging issue***
  Vapes and e-cigarettes are increasingly appearing in collection bins. Devices with nicotine or marijuana present added complexity, as they are classified as hazardous substances with costly disposal requirements even after battery removal.

- ***Needed support***
  Interviewees emphasized the value of a statewide public education campaign (modeled after Recycling Raccoons) and financial support for drop-off operations, including expanded producer responsibility (EPR) to cover all battery formats in Michigan.

CONSUMER BATTERY POLICY

Section Summary

This section summarizes current policy on consumer batteries in both Michigan and other states.

Key findings include:

- **Michigan gap:** Michigan has no statewide consumer battery program; current law only covers lead-acid and mercury restrictions.
- **Vermont model:** Vermont's EPR program (2014, expanding in 2026) shows how producer funding can ensure free and widespread statewide access and strong collection rates.
- **Other state action:** California (2022), Washington (2023), New Jersey & Illinois (2024), and Nebraska, Connecticut & Colorado (2025) have all passed battery EPR laws covering a range of battery types.
- **Harmonization trend:** Recent laws align on key provisions (e.g., limiting retailer obligations, prohibiting sales by non-compliant producers). Notably, Colorado requires a needs assessment for embedded batteries.
- **Path forward for Michigan:** Aligning future policy with these models would reduce industry opposition, streamline compliance, and expand convenient, consistent access.

Current Michigan Policy

Currently, Michigan's battery disposal law (Natural Resources and Environmental Protection Act Part 171) regulates the disposal of lead acid batteries, prohibits the sale of batteries containing intentionally introduced mercury, and provides guidelines for voluntary collection programs of nickel-cadmium batteries. Retailers that sell lead-acid batteries must accept lead-acid batteries from customers for recycling, which must be disposed of by a distributor, manufacturer, recycling or smelting facility (Michigan Legislature, 1994).

Individuals are encouraged to recycle all consumer batteries through retail collection or local drop-off locations, but there is no statewide program for consumer battery collection and recycling. Industry groups in Michigan, along with the Michigan Recycling Coalition (MRC), are working toward battery legislation. The following is a statement from Kerrin O'Brien, Executive Director of the MRC, on battery EPR:

The Michigan Recycling Coalition is committed to exploring extended producer responsibility as a policy approach to fund the recycling of a variety of end-of-life products. Currently the MRC is working with a potential legislative sponsor to draft a bill that outlines an extended producer responsibility (EPR) approach to the recovery and management of loose, small, and medium format batteries. The approach follows the latest state modeling of the EPR approach - battery collection and management that requires battery producers to assume financial and programmatic responsibility for their products. Stakeholders assume that actual collection and handling will be managed by interested public and private sector collection partners.

The Coalition is also exploring EPR for battery-embedded products, those products where the battery is not designed to be easily removed by the consumer. Material recovery facilities, haulers, fire fighters, and others across the country are reporting fires and thermal events when these batteries are damaged during transportation and processing. However, the enormity of the challenge has moved some states to include a report on batteries not included in current battery EPR laws in the coming years. Because of the number of industries that have batteries embedded in products, states will be working together to determine the best approach to managing these products at the end of their useful life. California is the only state with legislation around embedded batteries. Embedded batteries were added to the California Covered Electronic Waste Recycling Program which requires a fee collection at point of sale that supports the state managed recycling program for these products. California chose this approach because residents do not often recognize these products as battery-embedded but rather as electronic devices.

Best Practice Example: Vermont

In 2014, Vermont was the first state to implement an EPR law for single-use household batteries, or primary batteries, with the Vermont Primary Battery Stewardship Program. A Primary Battery is defined by the law as "a non-rechargeable battery weighing 4.4 pounds or less, including alkaline, silver oxide, zinc air, carbon-zinc, and lithium metal batteries" and excludes batteries that are not easily removeable or contained in medical devices (Vermont General Assembly, 2014). On January 1, 2026, batteries covered by the stewardship program will expand to include rechargeable batteries and the weight for covered batteries will increase to 25 pounds or less (Vermont General Assembly, 2024).

Vermont's battery EPR law requires producers of primary batteries sold in the state to register with the Vermont Department of Environmental Conservation and provide a stewardship plan for the proper recycling and disposal of primary batteries. This program is currently administered by the organization The Battery Network as a shared stewardship plan. This program operates at no cost to consumers, as it is fully funded by 24 producers of primary batteries sold in Vermont, who are designated as obligated stewards. In 2024, 99.97% of the state's population had access (living within a 15-mile radius) to a The Battery Network battery drop-off location. These battery drop-off sites are located at retailers, municipalities, and solid waste facilities. In 2024, Vermont collected 154,596 pounds of primary batteries, achieving a 29% collection rate. The program also collects rechargeable batteries, of which 76,749 pounds were collected in 2024 (Vermont Department of Environmental Conservation, 2024).

Other Examples

In 2022, California passed the Responsible Battery Recycling Act, which requires that producers of covered batteries establish a stewardship program that covers the collection and recycling of covered batteries in California. Covered batteries include loose primary and rechargeable batteries that are sold either separately from a product, that are designed to be easily removed from a product using common household tools, and batteries that are sold with, but not installed in, the product that the battery is designed to power. Covered batteries do not include primary batteries over two kilograms, rechargeable batteries over five kilograms, lead-acid batteries, motor vehicle batteries, fuel cell electrical generating facilities, batteries in medical devices, and recalled batteries (California Legislative Information, 2024). The stewardship program for battery collection and recycling is currently being developed by The Battery Network.

Other states have continued to pass policies regarding the collection and recycling of both single-use and rechargeable batteries. In 2023, Washington passed a battery EPR law that includes e-mobility device batteries (Product Stewardship Institute, 2023). In 2024, New Jersey and Illinois enacted new EPR laws for batteries: New Jersey's law also includes electric and hybrid vehicle batteries, while Illinois' law extends to medium-format batteries (MassRecycle, 2025). In 2025, Nebraska, Colorado, and Connecticut followed with EPR laws for portable and medium-format batteries, with Connecticut emphasizing fire risk reduction and public safety (Product Stewardship Institute, 2025).

Battery EPR Review

States enacting battery EPR legislation in 2025 have already achieved a notable degree of harmonization across several core program elements. These laws have consistent support from advocates such as the National Waste & Recycling Association and the Product Stewardship Institute, and they could be further strengthened by aligning key program design features. Harmonization reduces political friction, streamlines compliance for producers and retailers operating in multiple states, improves program efficiency, and supports consistent consumer education.

Recent laws in Nebraska (LB 36), Connecticut (HB 5019), and Colorado (SB 25-163) illustrate this trend. All three include similar retailer provisions (Table 12). Importantly, most new laws no longer require retailers to serve as collection sites for covered batteries—an approach that helps avoid significant opposition from the retail industry. This contrasts with Michigan, where retailers selling lead-acid batteries remain obligated to provide take-back service.

Beyond retailer obligations, battery EPR laws commonly address battery-containing products, though scope and definitions vary. For example, most statutes exclude medical devices and batteries that cannot be easily removed by users. Colorado's law goes further by requiring a statewide needs assessment to plan for end-of-life management of embedded batteries.

To maximize benefits, any new EPR laws is recommended to align with the frameworks recently adopted by other states. Greater harmonization can reduce political pushback, minimize industry opposition, and simplify compliance while ensuring consistent recycling access and messaging across jurisdictions.

Table 12: Statewide Battery EPR Laws Enacted in 2025

NE: LB 36	CT: HB 5019	CO: SB 25-163	RETAILER OBLIGATIONS
✓	✓	✓	Retailers may be captured in the definition of "producer"
✓	✓	✓	Retailers are not required to serve as collection sites for end-of-life batteries.
✓	✓	✓	Retailers may not sell covered products with improper labels
✓	✓	✓	Retailers may not sell covered products by non-compliant producers.
✓	✓	✓	Retailers may not charge a point-of-sale fee to consumers
	✓		Retailers are required to distribute educational materials to consumers
	✓		Retailers serving at collection sites are not required to take back damaged or defective batteries

CONSUMER BATTERY SOLUTION RECOMMENDATIONS

The following recommendations based on the findings in this report are potential options that EGLE may consider supporting:

Advisory group:
Establish an Advisory Group to guide battery recovery solutions and advance statewide strategies.

Needs assessment:
Assess existing recovery infrastructure and key challenges using the Battery Gap Analysis/Statewide Needs Assessment.

EPR legislation:
Advance EPR to shift costs to producers, align with other states, and secure stable funding; ensure coverage of embedded and emerging batteries (e.g., e-mobility, vapes, electronics) based on statewide needs assessment.

Program design:
Adopt design consistent with recent EPR laws to reduce opposition, simplify compliance, and support consistent consumer education; clarify retailer roles to avoid overburdening collection responsibilities.

Safety training:
Expanding training for public sector staff and collection organizations, along with stronger public education and outreach on battery hazards, to reduce battery fire hazard risks.

Access:
Guarantee year-round, statewide access through convenient drop-off options in both urban and rural areas.

Education:
Launch a statewide education and outreach campaign on battery recycling, leveraging Catalyst Communities and coordinated stakeholder engagement.

Processing capacity:
Expand recovery and processing capacity through targeted infrastructure investments supported by EGLE, federal funding opportunities, accelerators, and grant funding.

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APPENDIX

Residential Battery Survey

Response Demographics

A total of 776 Michigan residents participated in the residential battery recycling survey, representing 344 unique ZIP codes across the state. Responses were collected from all COGs, ensuring comprehensive geographic coverage. COG 8 had the highest participation, accounting for approximately 21% of total responses, while COG 11 had the lowest contribution, accounting for less than 1% of responses.

In terms of gender identity, 52% of respondents identified as female and 48% as male, with less than 1% identifying as non-binary or preferring not to disclose their gender. The age distribution of respondents showed that 42% were aged 44 or younger, 49% were between the ages of 45 and 74, and 10% were 75 or older. Respondents represented a wide range of ethnic backgrounds; however, 80% identified as White. The most represented demographic group overall was White women between the ages of 60 and 74, who accounted for 13% of total survey responses.

Survey Questions

The following are the survey questions administered to residents via the Qualtrics platform. Certain questions included survey logic, and both the questions and response options were often tailored based on the battery types provided by respondents. Notes on survey logic are included below in *blue*.

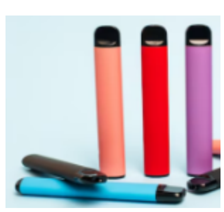
<u>Demographics</u>

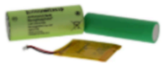
Q1. Are you 18 years or olde
Q2. What is your zip code?
Q3. What is your age range?
Q4. How would you describe your gender?
Q5. How would you describe your race or ethnicity?
Q6. Are you currently taking this survey from your home?

<u>Instructions Provided for Battery Identification</u>

Before beginning this survey, please familiarize yourself with the following battery types:

<table>
<tr><td></td><td>Single Use Batteries Alkaline and Zinc-Carbon Batteries - AAA, AA, C, D, 9V batteries common in toys, consumer electronics, medical devices</td></tr>
<tr><td>


</td><td>Single Use Lithium Batteries - AAA, AA, 9V, button-cell or coin batteries commonly found in watches, musical cards and books, string lights, digital scales and thermometers, cameras, smoke detectors, handheld games, remote controls, hearing aids, medical devices, keyless vehicle entry remotes, non-rechargeable e-cigarettes/vapes</td></tr>
</table>

      	**Rechargeable Lithium-Ion Batteries** - Cellphones such as iPhone and Android, laptops such as Dell and Macs, digital cameras, consumer electronics, newer cordless power tools such as Ryobi, DeWalt, Milwaukee, rechargeable e-cigarettes/vapes
  	**Rechargeable Nickel Cadmium Batteries** - Older cordless power tools, older digital cameras, and older bio-medical equipment
	**Rechargeable Nickle Metal Hydride Batteries** - Older type cellphones, digital cameras, cordless power tools, toys
	**Rechargeable Nickel-Zinc batteries** - Older type consumer electronics and medical devices

How do you know what type of battery your item is?

- Visual inspection of the item
    - Take battery cover off and look at battery type in the item
- Look up item online
    - Search for the item online and look for battery type information
- Estimate based on when the item was purchased. Rechargeable lithium-ion batteries have generally replaced rechargeable nickel chemistries such as nickel cadmium, nickel metal hydride, and nickel-zinc.
    - Laptops and cell phones transitioned to lithium-ion in 2005
    - Digital cameras transitioned to lithium-ion in 2006
    - Power tools transitioned to lithium-ion in 2010
    - Medical devices transitioned to lithium-ion in 2015
    - Kids toys and electric toothbrushes transitioned to lithium-ion in 2020

<u>Batteries Owned</u>

Q7. Which of the following types of batteries or battery-powered electronics do you currently have in your residence (either in use or in storage) *Check all that apply.*

- ☐ Single Use Batteries Alkaline and Zinc-Carbon Batteries
- ☐ Single Use Lithium Batteries
- ☐ Rechargeable Lithium-Ion Batteries
- ☐ Rechargeable Nickel Cadmium Batteries
- ☐ Rechargeable Nickle Metal Hydride Batteries
- ☐ Rechargeable Nickel-Zinc batteries

Q8. Approximately how many of each of the following batteries or battery-powered electronics do you currently have in your residence (in use or in storage)? *Battery options were carried forward based on the battery categories selected in Q7.*

- ☐ 1-2
- ☐ 3-5
- ☐ 6-10
- ☐ 10-20
- ☐ 20 or more

<u>Batteries Replaced</u>

Q9. In the past year, have you replaced any of the following batteries or battery-powered items from your residence? *Check all that apply.*

- ☐ Single Use Batteries Alkaline and Zinc-Carbon Batteries
- ☐ Single Use Lithium Batteries
- ☐ Rechargeable Lithium-Ion Batteries
- ☐ Rechargeable Nickel Cadmium Batteries
- ☐ Rechargeable Nickle Metal Hydride Batteries
- ☐ Rechargeable Nickel-Zinc batteries

Q10. If you selected any battery types above, about how many of each did you replace in the past year? *Battery options were carried forward based on the battery categories selected in Q9.*

- ☐ 1-2
- ☐ 3-5
- ☐ 6-10
- ☐ 10-20
- ☐ 20 or more

Q11. What did you do with the batteries you replaced? *Battery options were carried forward based on the battery categories selected in Q9.*

- ☐ Took them to a household hazardous waste (HHW) collection site
- ☐ Dropped them off at a local retailer (e.g., hardware or electronics store)
- ☐ Took them to a city or county recycling drop-off center
- ☐ Recycled through a mail-back program
- ☐ Placed in curbside trash bin
- ☐ Placed in curbside recycle bin
- ☐ Kept them at home / haven't gotten rid of them yet

Q12. If you still have the replaced batteries at home, please indicate why. *Battery options were carried forward based on the battery categories selected in Q9 if Kept at Home was selected in Q11.*

- ☐ Unsure how to recycle them
- ☐ Keeping the item for proper handling later
- ☐ Continuing to use the item in my current home (repurposed)

<u>Knowledge and Understanding</u>

Q13. How confident are you in your knowledge of what to do with each of these battery types when you're done using them? *Question asked for each battery type.*

- ☐ Very confident
- ☐ Somewhat confident
- ☐ Not very confident
- ☐ Not confident
- ☐ N/A, I don't use this type of battery

Q14. What do you believe is the proper way to dispose of the following types of batteries? *(Please select all options you believe are correct for each battery type)* *Question asked for each battery type.*

- ☐ Place in curbside trash bin
- ☐ Place in curbside recycle bin
- ☐ Recycle at special drop-off (municipal, retailer, etc.)
- ☐ Take to HHW site
- ☐ Recycle through a mail-back program

Q15. Are you aware that different types of batteries may require different handling methods?

- ☐ Yes
- ☐ No

Q16. Please indicate if you are aware of the battery handling requirements noted below. *Select yes / no for each statement.*

- ☐ Applying clear tape to the terminals of certain batteries (such as alkaline batteries larger than 9 volts, lithium, and lithium-ion) that have reached their end of life helps reduce the risk of fire.
- ☐ Damaged batteries require special handling, placing them in sand or kitty litter can help minimize the risk of fire.
- ☐ Embedded batteries are not intended to be removed separately; the entire device must be managed appropriately at end-of-life.
- ☐ It is best to store different battery chemistries separately.
- ☐ Avoid storing batteries in or near materials that conduct electricity, such as metal objects, as this can pose a safety hazard.
- ☐ Batteries need to be stored in a cool, dry area.
- ☐ Vapes and e-cigarettes may need to be emptied of nicotine to qualify for retailer take-back programs; otherwise, they are considered Household Hazardous Waste (HHW) due to their nicotine content.

Q17. Have you seen recycling instructions on battery packaging or labels?
- ☐ Yes, and I understood them
- ☐ Yes, but I did not understand them
- ☐ No
- ☐ Not sure

Barriers and Improvements

Q18. What, if anything, currently prevents you from recycling batteries? *Select all that apply.*
- ☐ I don't know where to take them for recycling
- ☐ It's inconvenient or difficult to access recycling sites
- ☐ I'm unsure which batteries can be recycled
- ☐ I don't have enough batteries to make recycling worthwhile
- ☐ I'm concerned about the cost of recycling
- ☐ I forget or don't think about recycling batteries
- ☐ I'm worried about safety or handling batteries
- ☐ There are no local recycling options available
- ☐ N/A; I do not have issues properly recycling my batteries

Q19. How easy or difficult is it for you to access battery recycling facilities or services in your community?
- ☐ Extremely easy
- ☐ Somewhat easy
- ☐ Somewhat difficult
- ☐ Extremely difficult
- ☐ I don't know where to find them or don't have access

Q20. Would you be more likely to recycle batteries if: *Select all that apply*
- ☐ There were more convenient drop-off locations nearby
- ☐ Recycling was available at local retail stores
- ☐ There were more community collection events for batteries
- ☐ I had clear information on how and where to recycle
- ☐ There were incentives or rewards for recycling batteries
- ☐ It was free and didn't cost anything to recycle
- ☐ N/A; I do not have issues properly recycling batteries

Q21. Have you ever been given recycling guidelines for batteries or battery-powered items from your county and/or community?
- ☐ Yes
- ☐ No
- ☐ I am not sure

Q22. If your community were to improve battery recycling, which of the following would you support? *Select all that apply.*
- ☐ More public education and outreach campaigns
- ☐ Partnerships with local stores to collect batteries
- ☐ Adding battery drop-off bins to public buildings (e.g., libraries, schools)
- ☐ Offering battery collection at annual clean-up or HHW events
- ☐ Providing safe storage containers for batteries at home
- ☐ Expanding types of batteries accepted at drop-off locations
- ☐ None of these options would be of interest to me

Q23. Do you have any suggestions or ideas for how your community could make battery recycling more convenient or effective? *Optional question.*

Q24. Are you interested in learning more about proper battery end-of-life management?

- ☐ Yes
- ☐ No

Q25. What topics would you like more information about? *Select all that apply. Question displayed if yes selected in Q24.*

- ☐ Which types of batteries can be recycled
- ☐ Where to take batteries for recycling
- ☐ Why proper battery management is important for safety and the environment
- ☐ Local programs or events that accept batteries
- ☐ How to identify battery types
- ☐ Not sure, just open to learning more

Q26. Please enter any other comments or suggestions you have about proper battery management in Michigan. *Optional question.*

Access searching methodology

The battery recycling programs researched represent approximately 80% of the population statewide. Program factors researched include the program website link, program name, the type of program (e.g. permanent drop-off operated by local government, temporary events operated by local government, private business, etc.), types of batteries accepted, who could utilize the program, and whether there was a fee to using the program. After the information was collected, a QA/QC protocol was followed:

- Review collected data and flag any inconsistencies or contradictory information.
- Re-search programs that were flagged during the review.
- Re-check data during final analysis to confirm consistency between attributes (e.g., service type vs. access type).
- Re-verify programs with high population counts for accuracy.
- Manually review programs with multi-family access or food waste access to ensure webpages and feedback notes accurately reflect coverage area and service type.